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MATLAB GUI Based Ultrasonic Liquid Level Monitoring System

In this project, an ultrasonic detector is mounted on top of a tank for the detection of liquid level. Monitoring the liquid level in tanks and vessels is one of the most important parameters in process industries.

In process plants, tanks contain

various liquids that are often quite expensive. Some of the liquids are inflammable and corrosive as well. Hence, it is better to monitor and control the level of liquid in tanks so that it does not overflow.

The ultrasonic level detector

pulse travels in air at a speed of 331m/sec at 0°C.

The authors' prototype is shown in Fig. 1. A screenshot of MATLAB based GUI used for level monitoring is shown in Fig. 2. The block diagram of the project is shown in Fig. 3.

Circuit and working

The major components used in this project are:

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller based development board with six analogue input pins and 14 digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI, and I2C. The microcontroller can operate at a clock frequency of 16MHz. In our project, digital I/O pins 11 and 12 of Arduino are connected to echo and trigger pins of the ultrasonic sensor module, respectively.

HC-SR04 ultrasonic sensor module. Ultrasonic ranging module HC-SR04 provides excellent non-contact range detection from 2cm to 400cm, with an accuracy of up to 3mm. The module includes an ultrasonic transmitter, receiver, and control circuit. The module used here has four pins—echo, trigger, Vcc, and GND. The basic principle of operation of the module is as follows:

Trigger pin is an input pin. It is made for a high pulse of at least 10µs duration, on receipt of which it automatically sends eight sonic bursts of 40kHz (ultrasonic signals). This signal, after reflecting from the object, is received by the echo pin.

The echo pin is an output pin that goes high for a period of time that is equal to the time taken for the ultrasonic wave to return to the sensor. The echo pin provides the travelling time of the sound wave in microseconds.

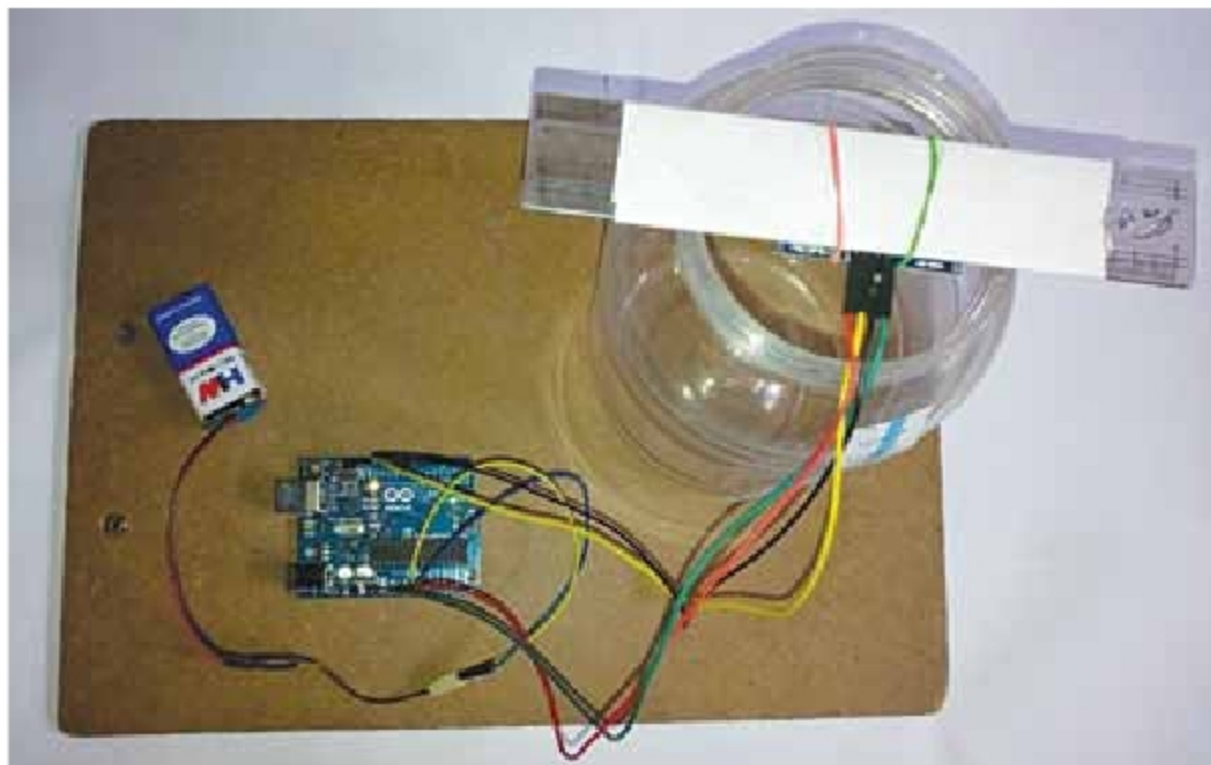


Fig. 1: Authors' prototype of the project

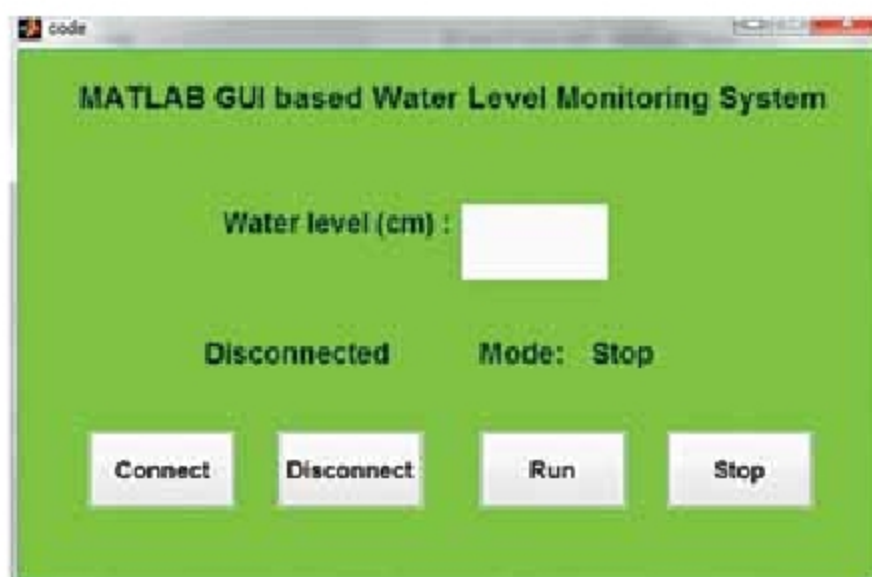


Fig. 2: Screenshot of MATLAB based GUI

proposed here is a popular non-contact type level detector. It operates by generating an ultrasonic pulse and measuring the time it takes for an echo to return. The travelling time indicates the depth of the empty space above the liquid in the tank. If the detector is mounted on top of the tank, the

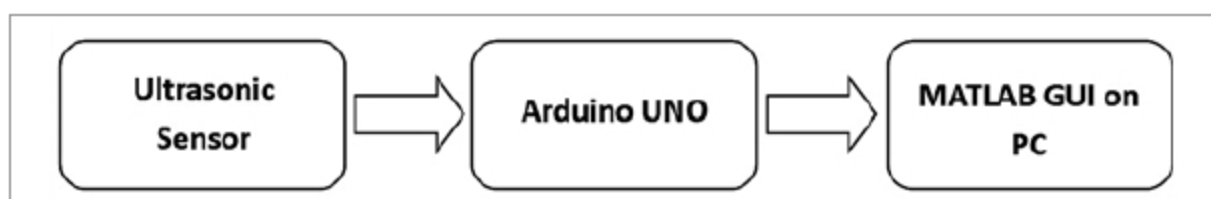


Fig. 3: Block diagram of the project